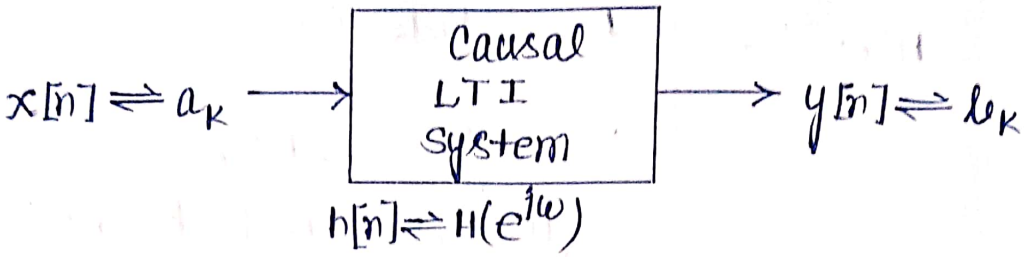
S&S QUIZ-6 SOLUTION

Sol(1):-



If $x[n]$ is periodic with 'N_0' then-

$$x[n] = \sum_{k=<N_0>} a_k \cdot e^{jk\omega_0 n} \quad \text{---- (1)}$$

therefore, $y[n] = x[n] * h[n]$

$$= \sum_{l=-\infty}^{\infty} x[n-l]\, h[l]$$

$$= \sum_{l=-\infty}^{\infty} h[l] \cdot \sum_{k=<N_0>} a_k \cdot e^{jk\omega_0(n-l)}$$

$$= \sum_{k=<N_0>} a_k \cdot e^{jk\omega_0 n} \cdot \sum_{l=-\infty}^{\infty} h[l] \cdot e^{-jk\omega_0 l}$$

$$= \sum_{k=<N_0>} a_k \cdot e^{jk\omega_0 n} \cdot H(e^{jk\omega_0})$$

$$= \sum_{k=<N_0>} a_k \cdot H(e^{jk\omega_0}) \cdot e^{jk\omega_0 n} \quad \text{---- (2)}$$

→ (1 Marks)

By eqn (2), Fourier series coefficient of $y[n]$,

$$b_k = a_k \cdot H(e^{jk\omega_0}) \quad \text{---- (3)}$$

& $$y[n] = \sum_{k=<N_0>} b_k \cdot e^{jk\omega_0 n} \quad \text{---- (4)}$$

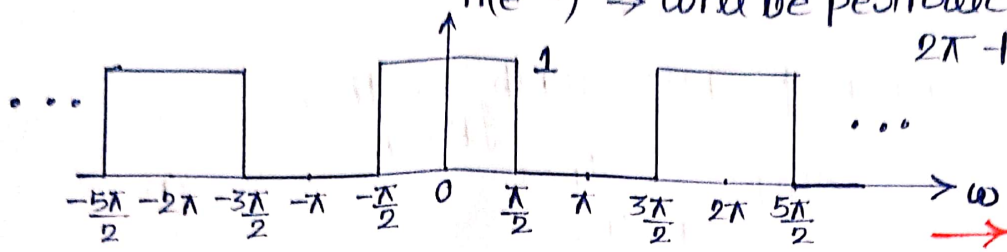
NOTE:- Input $x[n]$ & output $y[n]$ will be periodic with same time period.

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Given frequency response of LTI system,

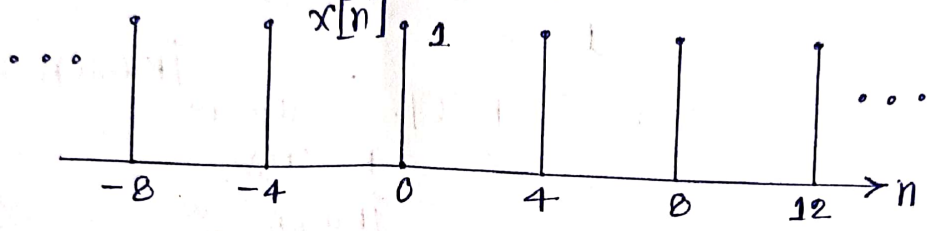
$$H(e^{j\omega}) = \begin{cases} 1 & |\omega| \leq \pi/2 \\ 0 & \pi/2 < |\omega| < \pi \end{cases}$$

$H(e^{j\omega})$ ⇒ will be periodic with 2π time period



→ (1.5 Mark)

Given input signal, $x[n] = \sum_{K=-\infty}^{\infty} \delta[n-4K] \rightleftharpoons a_K$



$$\therefore a_K = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] \cdot e^{-jk\omega_0 n}$$

Here, $N = 4$

$\omega_0 = \frac{2\pi}{N} = \frac{\pi}{2}$

$$= \frac{1}{4} \sum_{n=\langle 4 \rangle} x[n] \cdot e^{-jk\frac{\pi}{2}n}$$

$$= \frac{1}{4} \sum_{n=0}^{4} \delta[n] \cdot e^{-jk\cdot\frac{\pi}{2}n}$$

$$= \frac{1}{4} \quad \forall\ K$$

→ (1 Mark)

$$\therefore b_K = a_K \cdot H(e^{jk\omega_0}) \quad \& \quad b_K = b_{K+mN}$$

$$= \frac{1}{4} \cdot H\left(e^{jk\cdot\frac{\pi}{2}}\right)$$

$$b_0 = \frac{1}{4},\ b_1 = \frac{1}{4},\ b_2 = 0,\ b_3 = \frac{1}{4},\ b_4 = \frac{1}{4},\ b_5 = \frac{1}{4}, \ldots$$

$\therefore b_0 = b_4 = b_8 = b_{12} = \ldots = 1/4$

$\therefore b_1 = b_5 = b_9 = b_{13} = \ldots = 1/4$

$\therefore b_2 = b_6 = b_{10} = b_{14} = \ldots = 0$

→ (1.5 Marks)

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Sol(2):- Given signal, $x(t) = e^{-2|t-1|}$

$$\therefore X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} \cdot dt$$

$$= \int_{-\infty}^{\infty} e^{-2|t-1|} \cdot e^{-j\omega t} \cdot dt$$

$$= \int_{-\infty}^{1} e^{2(t-1)} \cdot e^{-j\omega t} \cdot dt + \int_{1}^{\infty} e^{-2(t-1)} \cdot e^{-j\omega t} \cdot dt$$

$$= \int_{-\infty}^{1} e^{-2} \cdot e^{(2-j\omega)t} \cdot dt + \int_{1}^{\infty} e^{2} \cdot e^{(-2-j\omega)t} \cdot dt$$

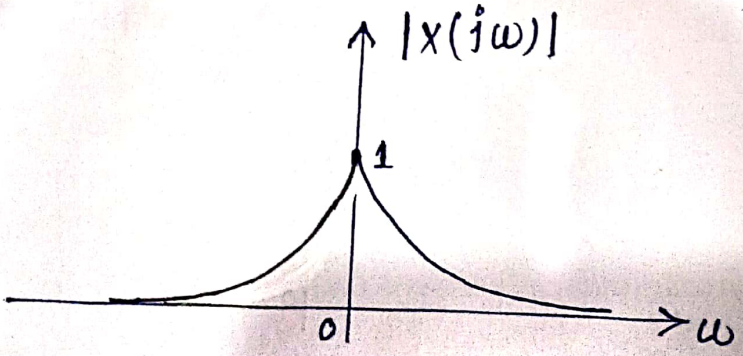
$$= e^{-2} \left\{ \frac{e^{(2-j\omega)t}}{(2-j\omega)} \right\}_{-\infty}^{1} + e^{2} \left\{ \frac{e^{(-2-j\omega)t}}{(-2-j\omega)} \right\}_{1}^{\infty}$$

$$= \frac{e^{-2}}{(2-j\omega)} \left\{ e^{(2-j\omega)} - 0 \right\} + \frac{e^{2}}{(-2-j\omega)} \left\{ 0 - e^{(-2-j\omega)} \right\}$$

$$= \frac{e^{-j\omega}}{(2-j\omega)} + \frac{e^{-j\omega}}{(2+j\omega)}$$

$$= \left(\frac{4e^{-j\omega}}{4+\omega^2} \right)$$

→ (3 Marks)



→ (2 Marks)

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